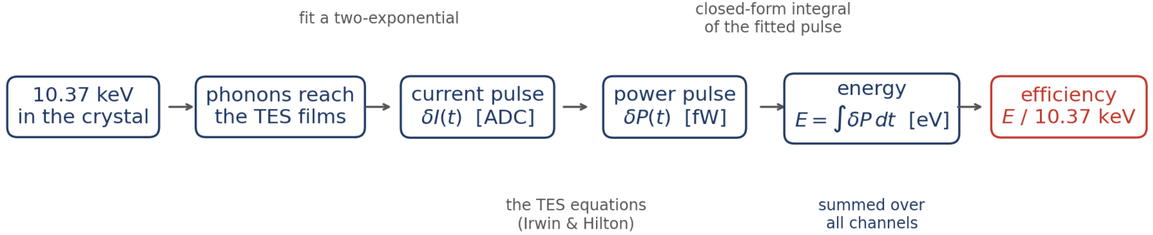
Phonon collection efficiency of Z7

Zhiheng Li · University of Minnesota · SuperCDMS SNOLAB Run 4 · September 2026
github.com/ZhihengLi0/urop → snolab_run4_phonon_pulse_analysis/collection_efficiency/

Z7 的声子收集效率

李志恒 · 明尼苏达大学 · SuperCDMS SNOLAB Run 4 · 2026 年 9 月
github.com/ZhihengLi0/urop → snolab_run4_phonon_pulse_analysis/collection_efficiency/

The Main Idea

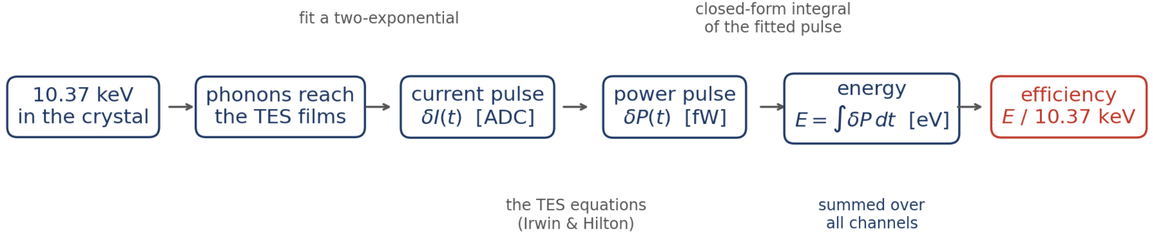


For detectors operated at 0 V (no NTL effect):

Collection efficiency = energy absorbed by the TESs, summed over all channels ÷ the 10.37 keV of the event

Sample: Z7, the K-line events of the Ge activation run, from SNOLAB R4 (2207 events, 30 series, all 11 channels).

主要思路



对于在 0 V 下工作的探测器（没有 NTL 效应）：

收集效率 = TES 吸收的能量（所有通道求和） ÷ 事件放进晶体的 10.37 keV

样本：Z7 ，Ge 活化的 K 线事件，来自 SNOLAB R4（ 2207 个事件， 30 个 series ， 11 个通道）。

The formula, and the two numbers it needs

Method 1 of the CDMS collection-efficiency note

$$\delta P(t) \;=\; I_0\,(R_L - R_0)\,\delta I(t) \;+\; 2\,R_L\,(\delta I(t))^2$$

linear term quadratic term

$$E \;=\; \int \delta P(t)\,dt \;=\; I_0(R_L - R_0)\,A(\tau_f - \tau_r) \;+\; 2R_L A^2\left(\frac{\tau_f+\tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f+\tau_r}\right)$$

for the fitted pulse $\delta I(t) = A\,[\,e^{-t/\tau_f} - e^{-t/\tau_r}\,]$ integrated from $-\infty$ to $+\infty$: zero before it starts, decays to zero after

Z7 PBS1: $I_0 = -18.07\ \mu$A, $R_0 = 44.16\ \text{m}\Omega$, $R_L = R_p + R_{sh} = 14.24\ \text{m}\Omega + 5.00\ \text{m}\Omega = 19.24\ \text{m}\Omega$

linear coefficient: $I_0(R_L - R_0) = (-18.07\ \mu\text{A}) \times (19.24\ \text{m}\Omega - 44.16\ \text{m}\Omega) = 0.45\ \mu\text{V}$

quadratic coefficient: $2R_L = 2 \times 19.24\ \text{m}\Omega = 38.5\ \text{m}\Omega = 0.0385\ \Omega$

δP = the change of Joule heating when the current moves by δI: a linear term and a small quadratic term.

I_0: TES current at the bias point · R_0: TES resistance at the bias point · R_L = R_p + R_sh: load resistance (R_p parasitic, R_sh shunt) · δI: change in current, from the trace

The note also gives Method 2: $\delta P = I_0(R_L - R_0)\cdot\delta I - R_L\cdot(\delta I)^2$ in our sign convention — same linear term, only the quadratic term differs, 1.7 % less energy (PBS1, median of 15 events).

Source: Antoine Rehberg, CDMS wiki, Collection Efficiency Analysis — confluence.slac.stanford.edu/spaces/CDMS/pages/542577254

公式和它需要的两个数

采用 CDMS 收集效率文档里的 Method 1

$$\delta P(t) \;=\; I_0\,(R_L - R_0)\,\delta I(t) \;+\; 2\,R_L\,(\delta I(t))^2$$

linear term quadratic term

$$E \;=\; \int \delta P(t)\,dt \;=\; I_0(R_L - R_0)\,A(\tau_f - \tau_r) \;+\; 2R_L A^2\left(\frac{\tau_f+\tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f+\tau_r}\right)$$

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quadratic coefficient: $2R_L = 2 \times 19.24\ \text{m}\Omega = 38.5\ \text{m}\Omega = 0.0385\ \Omega$

δP = 电流变化 δI 之后焦耳热的变化量：一个线性项，加一个很小的二次项。

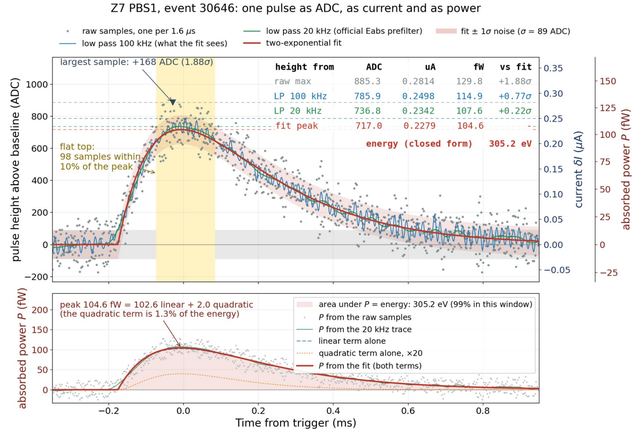
I_0 ：工作点上流过 TES 的电流 · R_0 ：工作点上 TES 的电阻 · R_L = R_p + R_sh ：负载电阻（ R_p 寄生电阻， R_sh 分流电阻 ）· δI ：电流的变化，来自波形

文档里还有一个 Method 2 ：换成我们的符号是 $\delta P = I_0(R_L - R_0)\cdot\delta I - R_L\cdot(\delta I)^2$ —— 线性项相同，只有二次项不同，能量低 1.7%（ PBS1 ， 15 个事件的中位数 ）。

来源： Antoine Rehberg 的 CDMS wiki 页面 Collection Efficiency Analysis — confluence.slac.stanford.edu/spaces/CDMS/pages/542577254

Example: one pulse

Z7 PBS1, event 30646: **the same trace as ADC counts, as current, as power**



baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA

$\delta P = I_0(R_L - R_0)\,\delta I + c_2 R_L\,(\delta I)^2$ (method1, c_2 = 2), with I_0 = -18.069 μA, R_0 = 44.159 mΩ, R_L = 19.240 mΩ

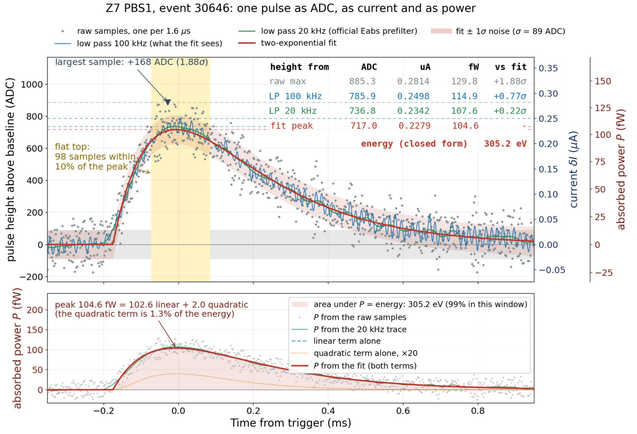
linear coefficient: $I_0(R_L - R_0)$ = (-18.069 μA) × (19.240 mΩ − 44.159 mΩ) = 0.450 μV

quadratic coefficient: $c_2 R_L$ = 2 × 19.240 mΩ = 38.48 mΩ = 0.0385 Ω

The two-exponential is fitted to the 100 kHz low-passed trace (blue), not to the raw samples; the raw samples are only drawn.

例子：一个脉冲

Z7 PBS1 ，事件 30646 ： **同一条波形，读成 ADC 计数、读成电流、读成功率**



baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA

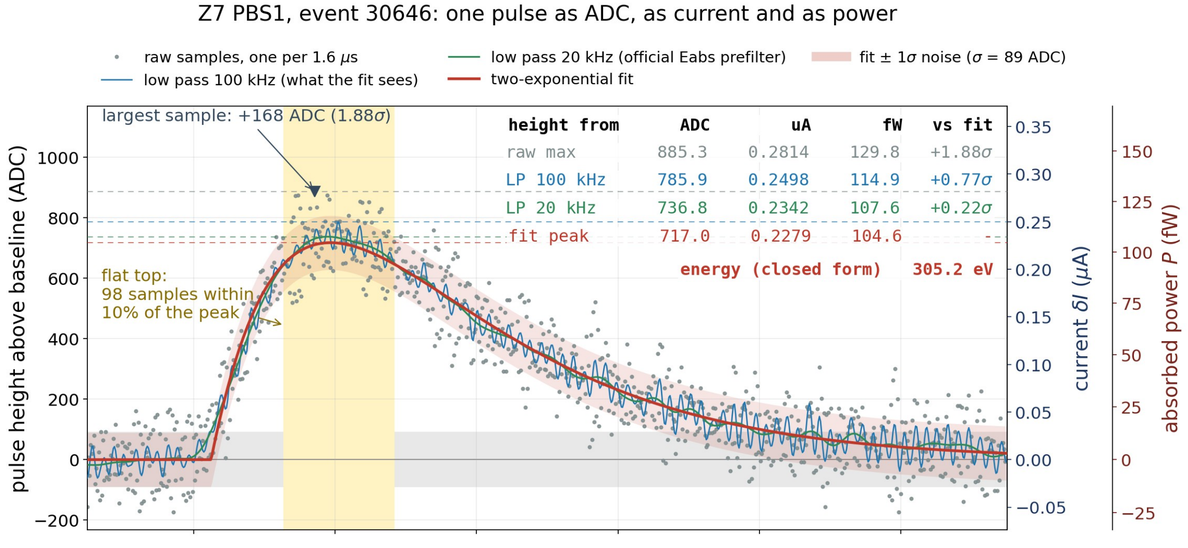
$\delta P = I_0(R_L - R_0)\,\delta I + c_2 R_L\,(\delta I)^2$ (method1, c_2 = 2), with I_0 = -18.069 μA, R_0 = 44.159 mΩ, R_L = 19.240 mΩ

linear coefficient: $I_0(R_L - R_0)$ = (-18.069 μA) × (19.240 mΩ − 44.159 mΩ) = 0.450 μV

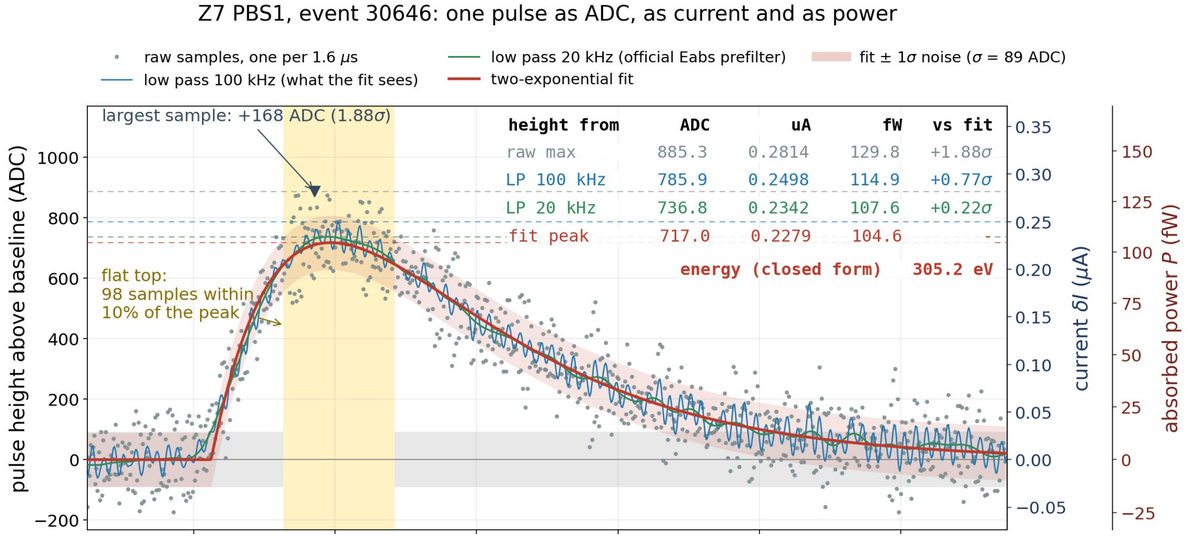
quadratic coefficient: $c_2 R_L$ = 2 × 19.240 mΩ = 38.48 mΩ = 0.0385 Ω

双指数拟合用的是 100 kHz 低通之后的波形（蓝线），不是原始采样点；原始采样点只是画出来对照。

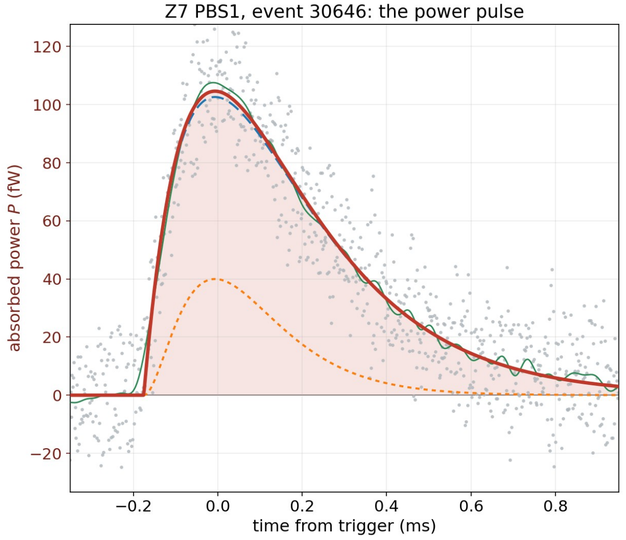
The height: take it from the fit, not from the largest sample



脉冲高度：从拟合取，不从最大的采样点取



The power pulse, and its area is the energy

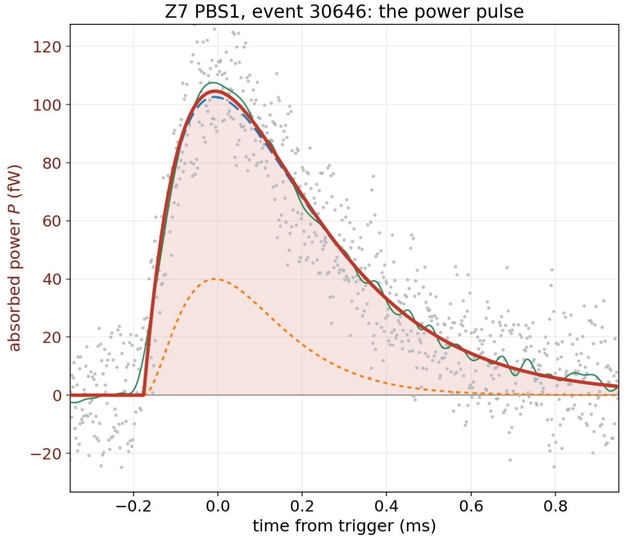


what is in the figure

- power from the fit, both terms
- the linear term alone
- the quadratic term alone, ×20
- from the raw samples, for comparison
- from the 20 kHz trace, for comparison
- area = the energy absorbed here

the quadratic term is 1.3 % of the energy,
so the power pulse has the shape of the cu

功率脉冲，它的面积就是能量

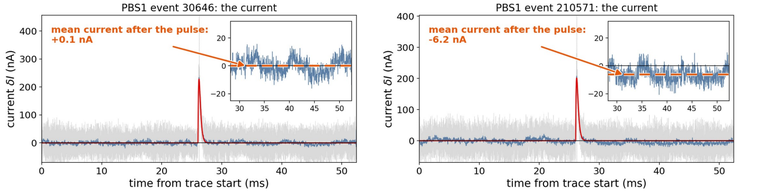


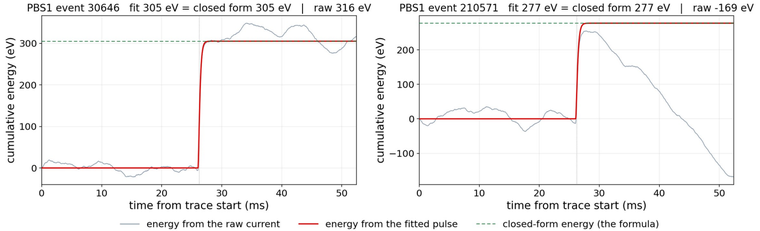
图里各是什么

- 拟合算出的功率，两项都算
- 只有线性项
- 只有二次项，放大 20 倍
- 用原始采样点算的，只作对照
- 用 20 kHz 波形算的，只作对照
- 面积就是这个通道吸收的能量

二次项只占能量的 **1.3%**，
所以功率脉冲的形状和电流脉冲一样

Why we integrate the fit, not the raw data

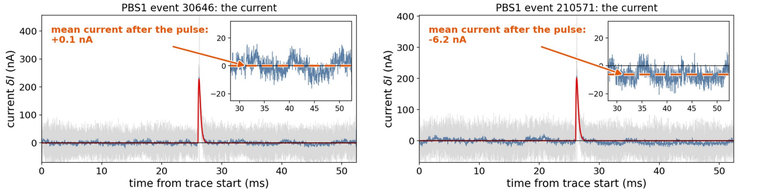


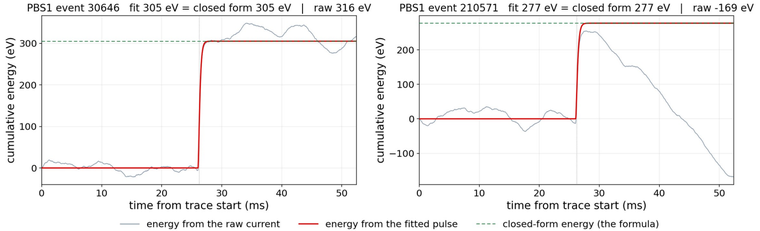


Top: the raw current of the two events; the small panel zooms in on the baseline after the pulse. Bottom: the energy accumulated from the raw current (grey) and from the fit (red).

Right: after the pulse the current sits 6 nA below its baseline — hidden in 28 nA of noise, yet over 24 ms it is −420 eV: raw −169 eV, fit 277 eV.

为什么对拟合积分，不是对原始数据



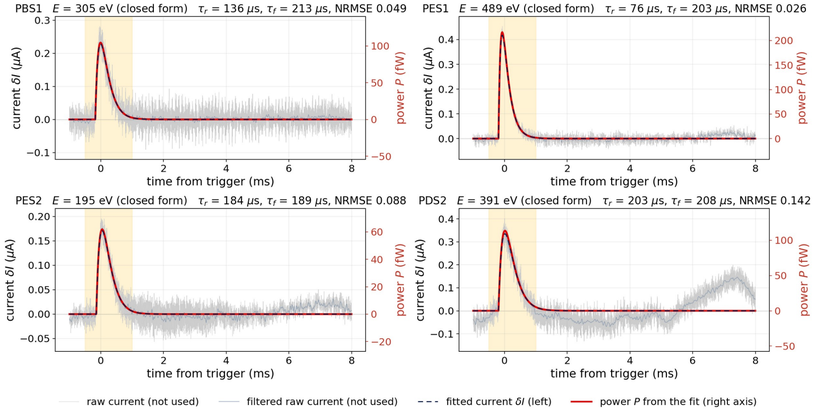


上：这两个事件的原始电流；右上角小图把脉冲之后的基线放大。下：由原始电流累计的能量（灰）和由拟合累计的能量（红）。

右：脉冲过后电流比基线低 6 nA —— 埋在 28 nA 的噪声里看不出来，但 24 ms 积下来就是 − 420 eV ：原始 − 169 eV ，拟合 277 eV。

The same event in its other channels: they add up to 33 %

Z7 event 30646; four of the eleven channels



energy per channel, event 30646

side 1	**2062 eV (60 %)**
PAS1	303 eV
PBS1	305 eV
PCS1	227 eV
PDS1	378 eV
PES1	489 eV
PFS1	359 eV
side 2	**1357 eV (40 %)**
PAS2	267 eV
PBS2	261 eV
PCS2	243 eV
PDS2	391 eV
PES2	195 eV

total 3419 eV = 33.0 % of 10.37 keV

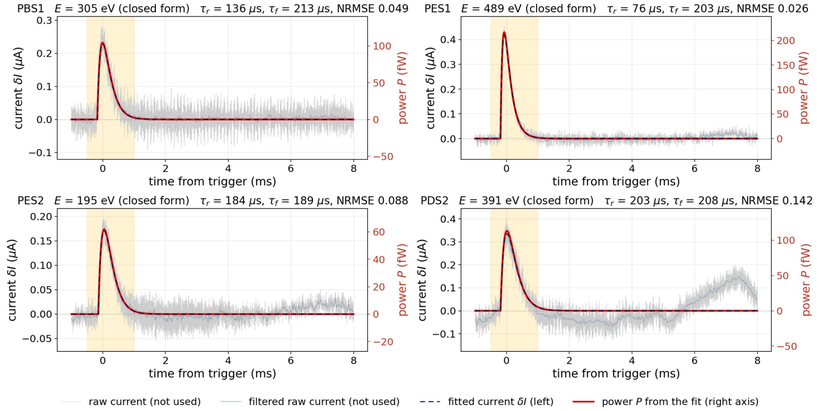
PFS2 is not read out on Z7, so side 2 has 5 channels
the channels also differ from each other; see the backup

The channels do not share the energy equally: 60 % of it lands on side 1, and PES1 alone has 489 eV against 195 eV in PES2.

Summed over the eleven channels: 3419 eV = 33.0 % of 10.37 keV. PDS2 (bottom right) carries a slow swing.

同一个事件的其他通道：加起来是 33%

Z7 事件 30646 ； 11 个通道里的 4 个



事件 **30646** 各通道的能量

第 1 面	**2062 eV (60 %)**
PAS1	303 eV
PBS1	305 eV
PCS1	227 eV
PDS1	378 eV
PES1	489 eV
PFS1	359 eV
第 2 面	**1357 eV (40 %)**
PAS2	267 eV
PBS2	261 eV
PCS2	243 eV
PDS2	391 eV
PES2	195 eV

合计 3419 eV = 33.0 % of 10.37 keV

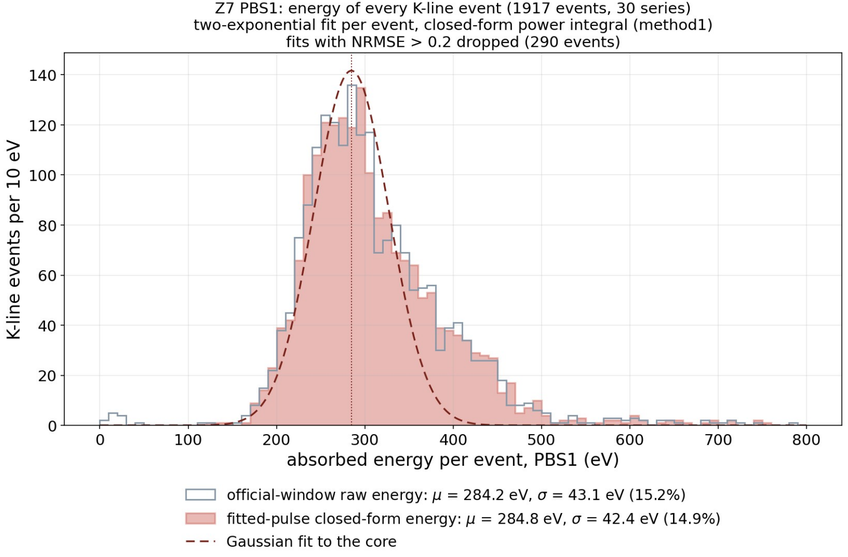
Z7 的 PFS2 没有读出，所以第 2 面只有 5 个通道
通道之间本身也有固定差别，见 backup 那一页

各通道分到的能量很不均匀： 60% 落在第 1 面； PES1 一个通道就有 489 eV ，而 PES2 只有 195 eV 。

11 个通道加起来： 3419 eV = 10.37 keV 的 33.0% 。 PDS2（右下）有一个缓慢的大起伏。

Now every event: one channel gives a wide, lopsided distribution

PBS1, 1917 K-line events, one fit and one integral each



Red: from the fit. Grey: the official window on the raw trace. Same peak to 0.2 %.

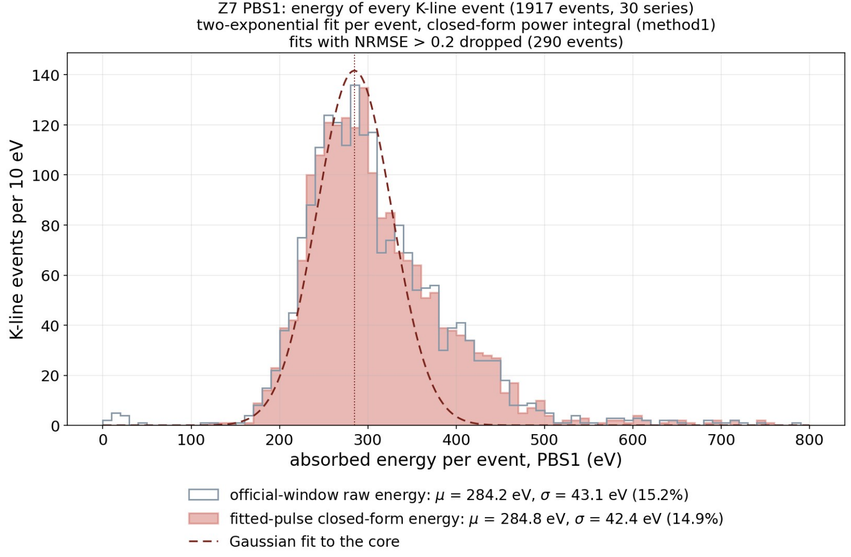
Peak 285 eV, width 15 % — for a line whose energy is fixed.

The tail to the right is the events that happened near this channel.

One channel's width is event position, not noise.

现在看所有事件：一个通道给出又宽又歪的分布

PBS1 ， 1917 个 K 线事件，每个事件一次拟合、一次积分



红色：来自拟合。灰色：官方窗口在原始波形上算的。峰位只差 0.2% 。

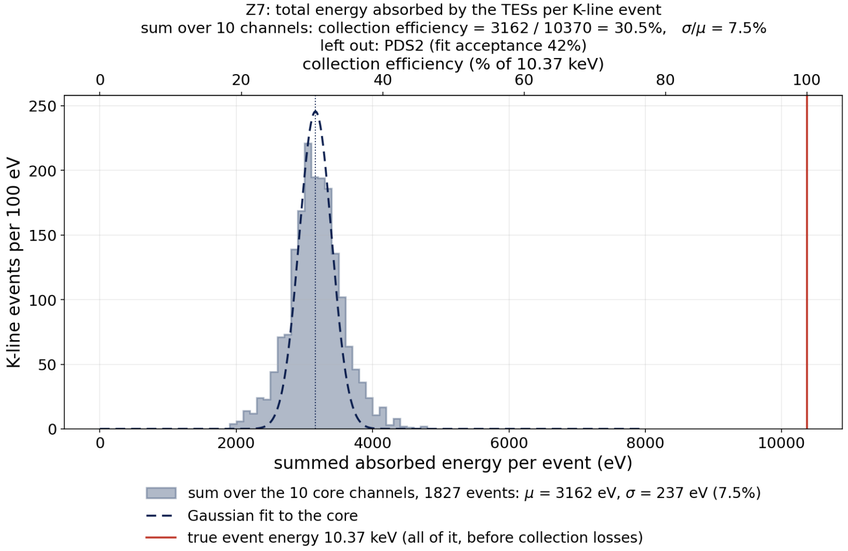
峰 285 eV ，宽度 15% —— 这可是一条能量固定的谱线。

往右拖的尾巴，是发生在这个通道附近的事件。

单通道的宽度是事件位置，不是噪声。

Summed over the channels: 30.5 %

the total energy the TESs absorbed per K-line event



Ten channels summed, 1827 events: 3162 eV = 30.5 %, and the width drops to 7.5 % — the position effect cancels.

PDS2 is left out: only 42 % of its events can be fitted, and those are the events that happened near it.

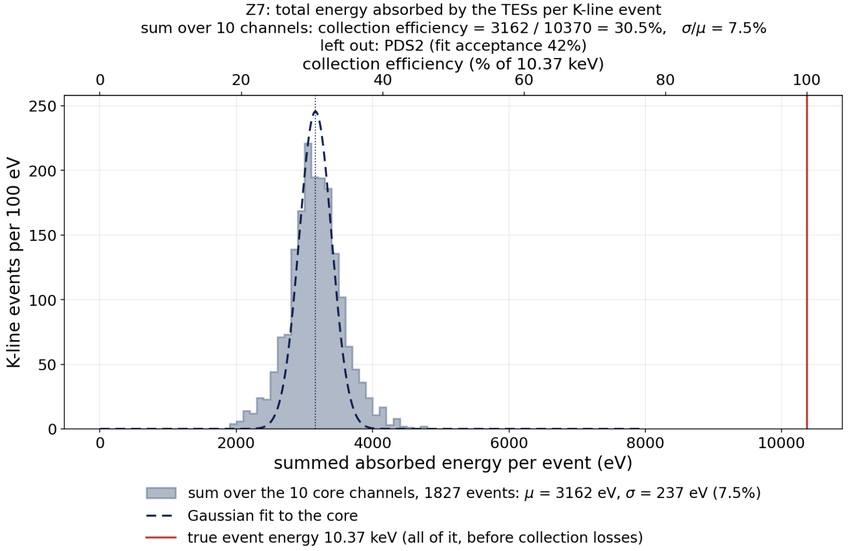
Collection efficiency:

30.5 % ± 2.3 % (σ/μ = 7.5 %).

Red line: 10.37 keV — two thirds of the energy never reaches the TESs.

各通道求和： 30.5%

每个 K 线事件被 TES 吸收的总能量



10 个通道求和， 1827 个事件： 3162 eV = 30.5% ，宽度降到 7.5% —— 位置效应抵消了。

PDS2 不算在内：它只有 42% 的事件能拟合上，而这些事件都发生在它附近。

收集效率 30.5% ± 2.3% （ σ/μ = 7.5% ）。

红线： 10.37 keV —— 三分之二的能量根本没到 TES 。

Summary, and what is still open

The current pulse becomes a power pulse through the TES equations; the area under the power is the energy.

The energy is taken from the fitted pulse, so the noise at the peak and the baseline drift do not enter.

Z7 absorbs 30.5 % ± 2.3 % (σ/μ = 7.5 %) of a 10.37 keV event in its TESs, summed over 10 channels. The CDMS note reports 26–41 % on the R37 CUTE tower.

Open: Z7 only · the bias of Z7 is not established (0 V assumed) · no dI/dV · PDS2 not included · 12.5 % of events fit in no channel.

总结，以及还没落实的

电流脉冲通过 TES 方程变成功率脉冲；功率下面的面积就是能量。

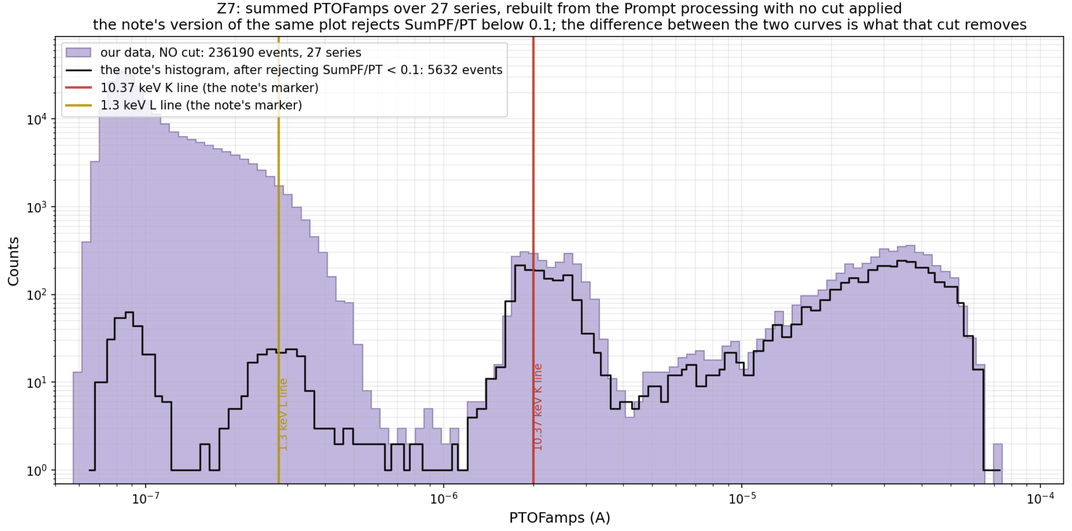
能量用拟合脉冲来算，所以不受峰值噪声和基线漂移的影响。

Z7 把一个 10.37 keV 事件的 30.5% ± 2.3%（σ/μ = 7.5% ）吸收进 TES ， 10 个通道求和。 CDMS 文档在 R37 CUTE tower 上得到的是 26–41% 。

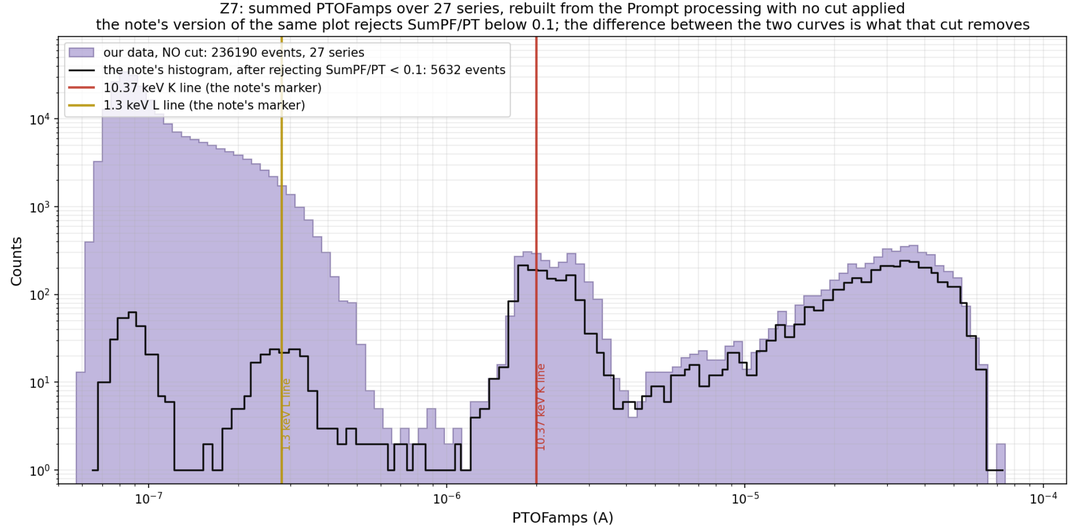
还没落实：只做了 Z7 · Z7 的偏压没有确认（按 0 V 算） · 没有 dI/dV · PDS2 没有算进去 · 12.5% 的事件在任何通道都拟合不上。

11

Backup — the events



Backup — 事件谱



Backup — the two Methods of the CDMS note

$\delta P = I_0(R_L - R_0)\cdot\delta I + c_2\cdot R_L\cdot(\delta I)^2$ with $c_2 = 2$ (Method 1, used here) and $c_2 = -1$ (Method 2)

The two differ only in the quadratic term: 1.7 % in energy (PBS1, median of 15 events) — far below the 15 % single-channel width and the 7.5 % width of the sum.

The inductor term of the full equations needs L and the loop gain from dI/dV, which do not exist for these series on MSI; it is neglected.

The official Eabs (trigger bin 16383, baseline bins 93..15758, 5-pole 20 kHz prefilter, window −0.5/+1 ms, same formula) is reproduced to 0.2 %; the grey outlines on the histogram slides are it.

Backup — 文档里的两个 Method

$\delta P = I_0(R_L - R_0)\cdot\delta I + c_2\cdot R_L\cdot(\delta I)^2$ $c_2 = 2$ 是 Method 1（本报告用的），$c_2 = -1$ 是 Method 2

两者只有二次项不同：能量差 1.7%（PBS1，15 个事件的中位数）—— 远小于单通道 15% 的宽度和求和后 7.5% 的宽度。

完整方程里的电感项需要 L 和 loop gain，要从 dI/dV 来，MSI 上这些 series 没有这个数据；这一项被忽略。

官方 Eabs（触发在第 16383 点，基线是 93..15758 点的平均，5 阶 20 kHz 预滤波，窗口 −0.5/+1 ms，同一个公式）复现到 0.2%；直方图页上的灰色轮廓就是它。

Backup — how the formulas are derived

1. From the current to the absorbed power

heat balance of the TES

$$C\dot{T} = P(t) + P_J - P_{bath}$$

over the whole pulse $\int C\dot{T}\,dt = 0$; the change in bath heat flow is not included

$$E = \int P\,dt = -\int \delta P_J\,dt$$

the circuit; at the bias point $V = I_0(R_L + R_0)$

$$V_{TES} = V - IR_L - L\dot{I}, \quad P_J = I\,V_{TES}$$

put $I = I_0 + \delta I$ and subtract the value at the bias point

$$\delta P_J = V\,\delta I - R_L\,(2I_0\,\delta I + (\delta I)^2) - L\,I\,\dot{\delta I}$$

integrate: the L term gives $L\,[I_0\,\delta I + (\delta I)^2/2]$, which is 0 at both ends

$$E = \int \delta P\,dt, \quad \delta P = I_0(R_L - R_0)\,\delta I + c_2\,R_L\,(\delta I)^2$$

c_2 = 1 from this derivation, 2 in Method 1 (used here), −1 in Method 2

2. Energy of the fitted pulse

the fit, with t counted from the pulse start t_0

$$\delta I(t) = A\,[\,e^{-t/\tau_f} - e^{-t/\tau_r}\,]$$

the only integral needed

$$\int_0^\infty e^{-t/\tau}\,dt = \tau$$

linear term

$$\int \delta I\,dt = A\,(\tau_f - \tau_r)$$

quadratic term: expand the square, integrate each exponential

$$\int (\delta I)^2 dt = A^2\left[\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r}\right]$$

so, with $c_2 = 2$

$$E = I_0(R_L - R_0)\,A(\tau_f - \tau_r) + 2R_L\,A^2\left[\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r}\right]$$

PBS1, event 30646: A = 1.39 μA, τ_r = 136 μs, τ_f = 213 μs ⇒ E = 305.2 eV

3. Symbols (Z7 PBS1 values) and assumptions

I_b: bias current R_{sh}: shunt resistor, 5.00 mΩ $V = I_b R_{sh}$: bias voltage, −1.146 μV R_p: parasitic resistance, 14.24 mΩ

$R_L = R_p + R_{sh}$: load resistance, 19.24 mΩ R_0: TES resistance at the bias point, 44.16 mΩ $I_0 = V/(R_L + R_0)$: TES current, −18.07 μA

L: inductance (unknown) δI: current change A: amplitude τ_r, τ_f: rise and fall time constants t_0: pulse start

Assumptions: detector at 0 V (no NTL effect); the heat flow to the bath is not included;

the L term cancels only after integration, so $\delta P(t)$ is not the exact power at each moment; only its integral is used

Backup — 公式推导和逻辑

1. 从电流到吸收的功率

TES 的热平衡

$$C\dot{T} = P(t) + P_J - P_{bath}$$

对整个脉冲积分，∫C·dT/dt dt = 0；不考虑流到热浴的热量的变化

$$E = \int P\,dt = -\int \delta P_J\,dt$$

电路方程；在工作点 V = I₀(R_L + R₀)

$$V_{TES} = V - IR_L - L\dot{I}, \quad P_J = I\,V_{TES}$$

代入 I = I₀ + δI，减去工作点的值

$$\delta P_J = V\,\delta I - R_L\,(2I_0\,\delta I + (\delta I)^2) - L\,I\,\dot{\delta I}$$

积分：电感项积出来是 L·[I₀·δI + (δI)²/2]，脉冲两端都是 0

$$E = \int \delta P\,dt, \quad \delta P = I_0(R_L - R_0)\,\delta I + c_2\,R_L\,(\delta I)^2$$

c₂ = 1 是这个推导的结果，2 是 Method 1（本报告用），−1 是 Method 2

2. 拟合脉冲的能量

拟合函数，t 从脉冲开始的时刻 t₀ 算起

$$\delta I(t) = A\,[\,e^{-t/\tau_f} - e^{-t/\tau_r}\,]$$

只需要这一个积分

$$\int_0^\infty e^{-t/\tau}\,dt = \tau$$

线性项

$$\int \delta I\,dt = A\,(\tau_f - \tau_r)$$

二次项：把平方展开，每个指数分别积分

$$\int (\delta I)^2 dt = A^2\left[\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r}\right]$$

所以，取 c₂ = 2

$$E = I_0(R_L - R_0)\,A(\tau_f - \tau_r) + 2R_L\,A^2\left[\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r}\right]$$

PBS1，事件 30646：A = 1.39 μA，τr = 136 μs，τf = 213 μs ⇒ E = 305.2 eV

3. 符号（Z7 PBS1 的数值）和假设

I_b：偏置电流 R_sh：分流电阻，5.00 mΩ V = I_b·R_sh：偏置电压，−1.146 μV R_p：寄生电阻，14.24 mΩ

R_L = R_p + R_sh：负载电阻，19.24 mΩ R₀：工作点的 TES 电阻，44.16 mΩ I₀ = V / (R_L + R₀)：工作点的 TES 电流，−18.07 μA

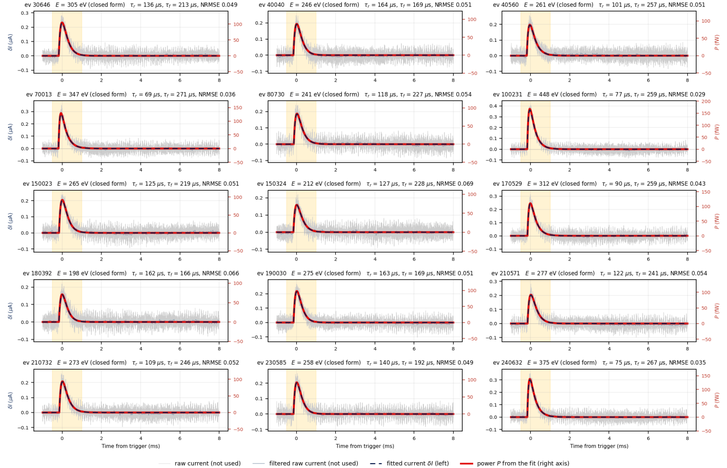
L：电感（未知） δI：电流变化 A：幅度 τr、τf：上升和下降时间常数 t₀：脉冲开始的时刻

假设：探测器在 0 V（没有 NTL 效应）；不包括流到热浴的热量；

电感项只有积分以后才为 0，所以 δP(t) 不是每一时刻的精确功率，我们只用它的积分

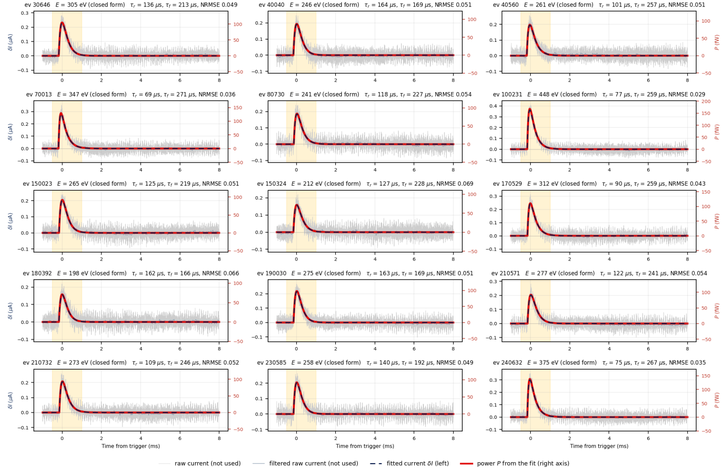
Backup — 15 events, fitted current and power (PBS1)

Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)
Z7 24260617_063934, K-line events | method1: P(t) = I0(R_L-R0)*dI(t) +2*R_L*(dI(t))^2 | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model A(exp(-t/tau_f)-exp(-t/tau_r)), 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



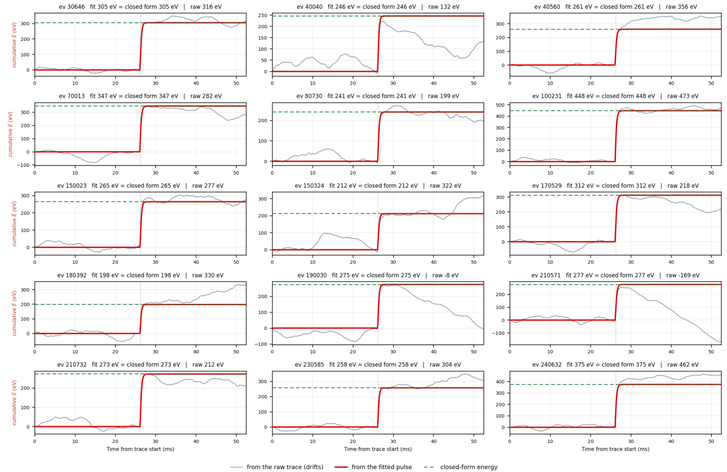
Backup — 15 个事件，拟合电流和功率（ PBS1 ）

Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)
Z7 24260617_063934, K-line events | method1: P(t) = I0(R_L-R0)*dI(t) +2*R_L*(dI(t))^2 | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model A(exp(-t/tau_f)-exp(-t/tau_r)), 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



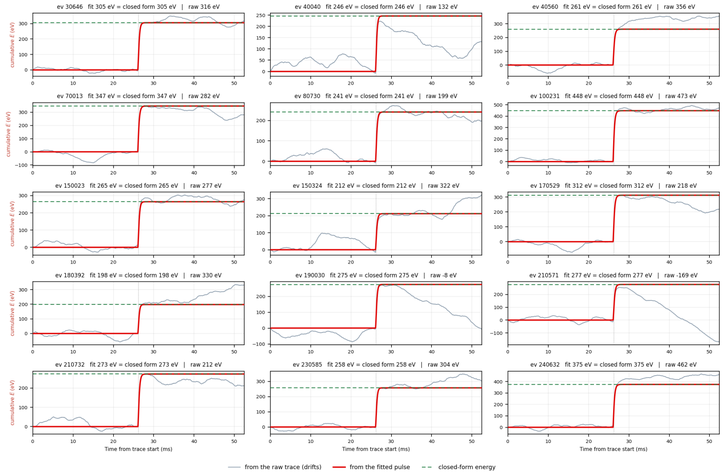
Backup — 15 events, cumulative energy (PBS1)

Z7 PBS1: integrated power = absorbed energy, fitted pulse vs raw trace
Z7 24260617_063934, K-line events | method1: P(t) = I0(R_L·R0²dI(t) +2*R_L*(dI(t))^2 | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model A(exp(-(tau, f)-exp(-(tau, r)], 5 free params, pretrigger free in 16383+3000); the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window

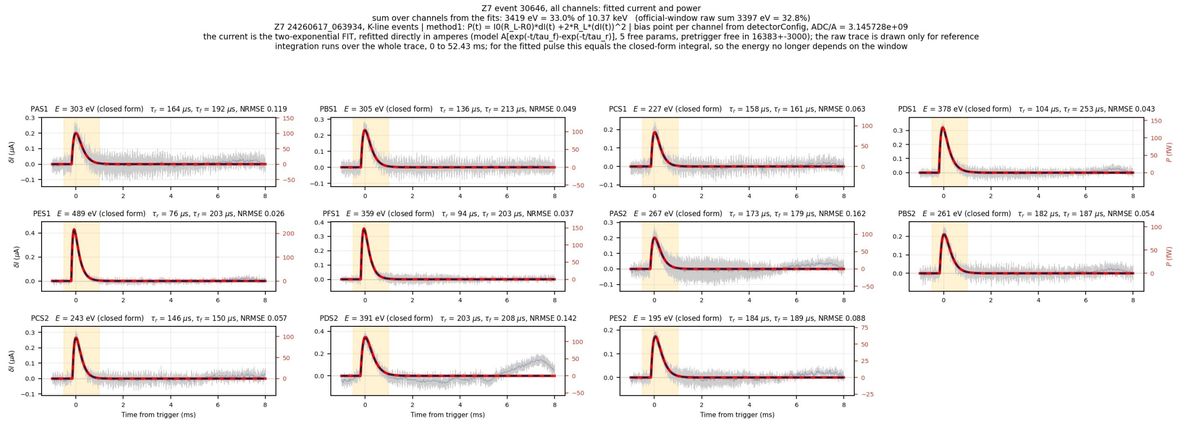


Backup — 15 个事件，累计能量（ PBS1 ）

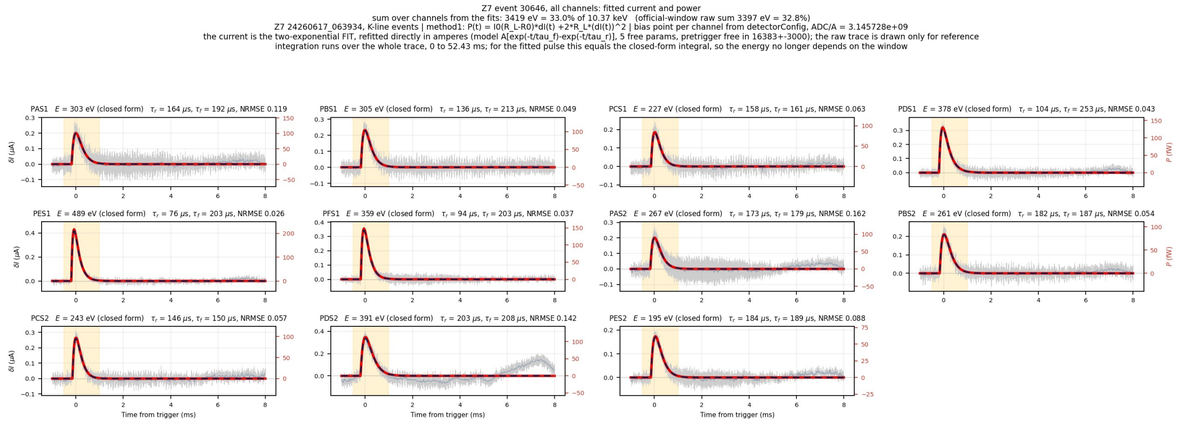
Z7 PBS1: integrated power = absorbed energy, fitted pulse vs raw trace
Z7 24260617_063934, K-line events | method1: P(t) = I0(R_L·R0²dI(t) +2*R_L*(dI(t))^2 | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model A(exp(-(tau, f)-exp(-(tau, r)], 5 free params, pretrigger free in 16383+3000); the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



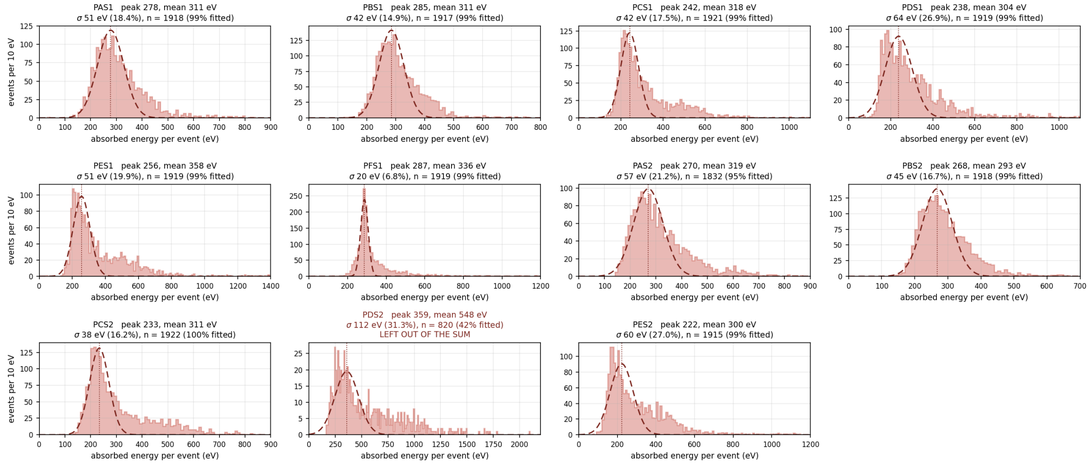
Backup — event 30646, all channels



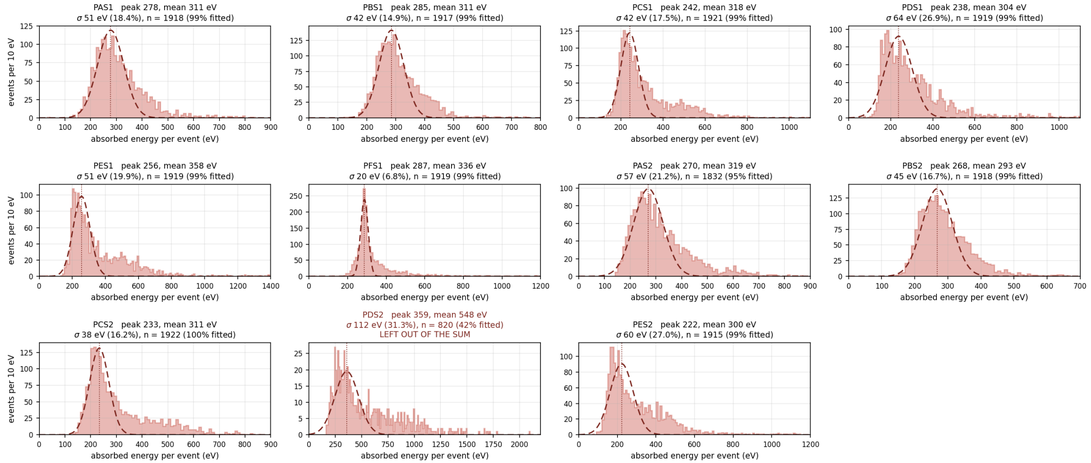
Backup — 事件 30646 ，全部通道



Backup — energy distribution of every channel



Backup — 每个通道的能量分布



Backup — event bookkeeping: what is excluded at each step

PTOFamps window around the K line (30 series, no trigger cut): 2207 events

− 276 events (12.5 %): the two-exponential fit fails in every channel (all NRMSE > 0.2) — pileup / glitches, not checked one by one

= 1931 events with at least one usable fit

− 104 events: at least one of the 10 core channels failed to fit — an incomplete sum would read low

= 1827 events (83 % of 2207) enter the summed-energy distribution

Channel-level cuts, applied separately: PDS2 left out (fits only 42 % of events, the ones near it) · PFS2 has no data on Z7

Backup — 事件计数：每一步排除了多少

K 线附近的 PTOFamps 窗口（ 30 个 series ，无 trigger cut ）： 2207 个事件

– 276 个（ 12.5% ）：双指数拟合在每个通道上都失败（ NRMSE 全部 > 0.2 ）—— 疑似 pileup / glitch ，未逐个检查

= 1931 个至少有一个通道拟合成功

– 104 个： 10 个核心通道里至少有一个拟合失败 —— 总能量缺一块，加进去会偏低

= 1827 个（占 2207 的 83% ）进入总能量分布

通道层面的排除（另算）： PDS2 不计入（只有 42% 的事件能拟合，而且都是它附近的事件） · PFS2 在 Z7 上没有数据